PHASOR: Phase-Aware Sparse Attention with Learned Orthogonal Routing

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Abstract

Sparse attention methods decide which key–value pairs a query may see. Two families dominate: *hash-based* methods, which bucket tokens by a random angular projection (Reformer), and *cluster-based* methods, which bucket them by learned centroids (Routing Transformer). We revisit this design space from the geometry of the router itself, propose an architecture built on three choices, and then subject each of them to controlled numerical study against a reference implementation.

The architecture, PHASOR, routes tokens by a *learned orthogonal* rotation acting on a *fixed* orthonormal codebook; reads a routing subspace that rotary position embeddings do not touch; and makes the tokens a query's bucket excludes visible to the softmax as coarse residue summaries rather than skipping them.

The studies are the substance of this report, and two of them overturned claims we had made. We show that (i) the codebook geometry is provably rigid under training, which an unconstrained router's is not — we drive a free router's minimum prototype angle from 48.4° to 0° while ours stays at exactly 90°; (ii) the bucket load is a function of the *input distribution* and not of the router's parameters, so a balancing loss placed on the router cannot work, which invalidated one of our own planned ablations; (iii) rotary phase does not degrade angular routing toward uniform hashing as we had claimed, but toward a hard *capacity ceiling* of five to ten effective buckets that is nearly independent of how many are provisioned — a worse failure, with a geometric explanation: the routing input's trajectory over positions is confined to a torus of dimension $d_c/2$ and therefore explores a measure-zero subset of the sphere; and (iv) the bucket-sorted chunking that the method's $O(n^{1.5})$ complexity depends on is faithful only when the chunk size is comparable to the mean bucket size, retaining as little as 38% of true in-bucket pairs otherwise, which couples the routing width to the context length and weakens the complexity claim.

We report no language-modelling results, and every table marked as such is a placeholder. What we do report is reproducible from the released code with a single command. We state precisely which components are novel, which are recombinations of published ideas, which of our own claims the evidence has already corrected, and which predictions would falsify what remains.

1 Introduction

The attention mechanism [40] computes, for every query, a distribution over every key. This is the operation that makes transformers general and the operation that makes them expensive. A long line of work reduces the cost by restricting *which* keys each query sees: by fixed patterns [4, 10, 43], by random angular hashing [22], by online clustering [34], or by learned block selection [26, 42]. A parallel line reduces cost by restricting *which parameters* each token uses, through mixture-of-experts routing in the feed-forward layer [17, 23, 36], in the attention projections [13, 21, 44], or across depth [32].

These two lines share a component that receives comparatively little design attention: the **router**. In most systems the router is a single linear map followed by a softmax or a top-k, trained end to end with an auxiliary load-balancing loss. This is known to be brittle. Chi et al. [8] show that routing through a low-rank linear map drives expert prototypes toward one another — *representation collapse* — and propose a cosine-similarity router with L_2-normalised prototypes as a remedy. Zoph et al. [46] document the stability problems that motivate router z-losses. Roller et al. [33] show that a *fixed random hash* is a competitive router, which is an uncomfortable result for learned routing: it suggests that much of what learned routers contribute is partitioning, not semantics.

This report takes the geometry of the router as its primary object of study. Our starting observation is that the two most successful content-based sparse-attention mechanisms are both *angular*. Reformer assigns buckets by arg max over $[\mathbf{xR}; -\mathbf{xR}]$ for a *random* rotation $\mathbf{R}$ — cross-polytope LSH [2, 7]. Routing Transformer runs spherical k-means on L_2-normalised queries and keys, so its centroids are also directions. In both cases the quantity that determines routing is an angle. The difference is only whether the frame defining the sectors is random and fixed (Reformer) or data-fit and learned (Routing Transformer).

That framing suggests the intermediate design we study here. Let the frame be *learned*, as in Routing Transformer, but constrained to be *orthogonal*, as in Reformer. The router applies a learned rotation $\mathbf{R} \in SO(d_c)$ to a normalised token representation and assigns the bucket by arg max over a *fixed* orthonormal codebook. We call the resulting attention layer **PHASOR** (Phase-Aware Sparse attention with Orthogonal Routing).

1.1 The three design choices

A constrained frame, not a free one. If the codebook were also learned, the rotation would be redundant — $\langle \mathbf{R}\hat{\mathbf{u}}, \mathbf{c}_b \rangle = \langle \hat{\mathbf{u}}, \mathbf{R}^\top \mathbf{c}_b \rangle$ — and we would have reinvented a cosine router. Because the codebook is fixed and orthonormal and the learned map is constrained to $O(d_c)$, the *geometry of the partition* is invariant under training: sectors stay equiangular and mutually separated by $\pi/2$ no matter what the optimiser does (Proposition 1, measured in Section 5.4). Training moves the *data* relative to a rigid partition rather than moving a deformable partition relative to the data.

Routing that does not read rotary dimensions. Reformer and Routing Transformer were designed before rotary position embeddings [37] became the default. RoPE encodes position by *rotating* query and key subvectors as a function of position, so an angular router reading RoPE-rotated features reads a phase that mixes content and position. We make this precise in Proposition 2 and measure it in Section 5.2. The fix is structural and cheap: route on a subspace that RoPE does not touch. Partial application of RoPE is already standard practice for unrelated reasons [6], and decoupled RoPE dimensions are used in DeepSeek-V2's latent attention [16]; we give the routing-specific argument for it.

A fallback the softmax can see. In Mixture-of-Depths [32] and in capacity-overflow handling in MoE layers [17, 23], an unselected token is carried forward by the residual stream and is otherwise *absent* from the block. Information is preserved, but the excluded token is not merely down-weighted — it is not present in the softmax at all, so attention cannot express "I looked and it was irrelevant" as distinct from "I was not allowed to look". We propose instead that the router's residue be made visible at low resolution: the tokens a query's bucket excludes are pooled into B summary key–value pairs and appended to that query's key set, at $O(nB)$ cost. This is related to, but distinct from, global tokens [4, 43], attention sinks [41], registers [15], and NSA's compression branch [42]: in all of those the coarse branch is *independent* of the routing decision, whereas here the summaries are exactly the complement of what the router selected, and therefore carry a different signal for each query.

1.2 What this report adds beyond the design

A design is cheap. We therefore built the reference implementation first and measured each claim before asserting it. That process changed the paper four times, and we report the changes rather than the polished end state, because the corrections are more informative than the original claims were.

1. **Proposition 4, which we did not expect.** The bucket marginal under an orthogonal router with a fixed codebook is *invariant to the router's parameters* whenever the routing input is isotropic. This is a stronger guarantee than Proposition 1 and it is free. It also has a cost: it means a balancing loss whose gradient reaches only **R** cannot change the load, which **invalidated ablation A6 as we had specified it** (Section 5.3).

2. **A capacity ceiling, not uniform hashing.** We had claimed that routing on RoPE dimensions degenerates toward chance, $1/B$. Measurement shows the aggregate limit is roughly 0.14–0.20 co-routing agreement regardless of B, because the router reaches only five to ten *effective* buckets however many are provisioned — 37% of a 16-bucket codebook but only 8% of a 128-bucket one. Proposition 3 explains why: the routing input's trajectory over positions lies on a torus of dimension $d_c/2$ inside a d_c-dimensional sphere, and therefore meets only a vanishing fraction of the partition's cells. The practical consequence is the opposite of reassuring: provisioning a larger codebook makes the relative collapse worse (Section 5.2).

3. **The complexity claim is weaker than stated.** The $O(n^{1.5})$ figure assumes bucket-sorted fixed-size chunks. We measure how far that approximation is from true bucket-wise attention and find it is faithful only when the chunk size is comparable to the mean bucket size n/B. At $n = 4096$ with $B = 32$ and chunk 32, the chunked pattern retains 38% of true in-bucket pairs, with load entropy at 0.996 — so this is not a load-imbalance artefact. Since $B = 2d_c$ is tied to the head's content width, holding fidelity fixed as n grows requires d_c to grow with $\sqrt{n}$ (Section 5.7).

4. **A null result that constrains the protocol.** At initialisation, all four ablation arms produce patterns of indistinguishable quality — recall-over-density between 2.5 and 4.7 for every variant, on both unstructured and content-structured inputs. The cross-polytope partition beats a random pattern of equal density by a factor of three to five, but *which* angular rule is used makes no measurable difference before training. Ablations A1 and A3 therefore cannot be settled by pattern geometry; they require training (Section 5.5).

What this report is and is not. This is a method-and-evidence report, not a results paper. The components we build on are published, and Section 7 maps each element of our design onto its closest antecedent, including cases where the antecedent is very close. We have trained no language model; every table of language-modelling results is a placeholder, marked as such. The evidence we do present is of one kind — controlled measurement of the mechanism on constructed inputs against the reference implementation — and Section 5.1 states exactly what that kind of evidence can and cannot support. Section 6 specifies the experiments that would settle the rest, revised in light of what the studies found, including the outcomes that would falsify each remaining claim.

2 Background and Related Work

2.1 Angular routing in sparse attention

Reformer. Kitaev et al. [22] share the query and key projections ($\mathbf{q}_i = \mathbf{k}_i$) and assign each token a bucket by locality-sensitive hashing: draw a random matrix $\mathbf{R} \in \mathbb{R}^{d \times b/2}$ and set $h(\mathbf{x}) = \arg\max([\mathbf{xR}; -\mathbf{xR}])$. This is cross-polytope LSH [2]: the sign-and-argmax structure partitions the sphere into b sectors whose boundaries are determined by a random orthogonal frame. Attention is computed only within buckets, and n_r independent hash rounds reduce the probability that two nearby vectors are separated. Cost is $O(n \log n)$ with chunked sorting.

Two properties matter for us. First, $\mathbf{R}$ is *random and fixed*: the partition is content-agnostic and its quality is a statistical guarantee, not a learned one. Second, the codebook is the cross-polytope $\{\pm\mathbf{e}_j\}$, i.e. orthonormal. PHASOR keeps the second property and learns the first.

Routing Transformer. Roy et al. [34] cluster L_2-normalised queries and keys with online spherical k-means and let each query attend within its cluster, giving $O(n^{1.5})$ cost. The centroids are learned (by EMA, not by gradient descent through the assignment). Because centroids are unconstrained directions, nothing prevents them from concentrating; the method relies on k-means dynamics and on balanced top-k assignment for stability.

Learned block sparsity. NSA [42] and MoBA [26] select *blocks* of keys per query using pooled block representations, and are designed for hardware efficiency and native trainability. NSA in particular combines three branches: compressed coarse-grained tokens, selected fine-grained blocks, and a sliding window. Our residue-summary fallback resembles NSA's compression branch in shape and differs in construction, as noted above. These two systems are the strongest recent baselines for any trainable sparse attention proposal, and we include them in the protocol.

2.2 Routing in mixture-of-experts layers

Sparse MoE routing [17, 23, 36] selects parameters rather than keys, but the router design problems are the same. Chi et al. [8] attribute instability to representation collapse induced by routing through a low-rank linear projection, and propose routing by cosine similarity against L_2-normalised expert embeddings in a low-dimensional space. That method (X-MoE) is the closest published relative of our router: both normalise, both score by angle. The difference is that X-MoE's prototypes are free parameters and can in principle still concentrate, whereas our prototypes are a fixed orthonormal frame and the learned degrees of freedom live entirely in $O(d_c)$.

Lewis et al. [24] cast assignment as a balanced linear assignment problem, guaranteeing balance by construction at the cost of a global solve. Roller et al. [33] use fixed random hashes and show they are competitive — a result we treat as the null hypothesis that any learned router must beat. Clark et al. [11] give scaling laws for routed models. Jiang et al. [20] report that expert specialisation in a strong MoE model appears largely syntactic and positional rather than semantic, which is relevant context for any proposal motivated by semantic specialisation, including this one (Section 4.4).

2.3 MoE inside attention

MoA [44], SwitchHead [13] and MoH [21] place experts inside the attention layer. SwitchHead's ablation is directly informative for the present design: *the best configuration places experts on the value and output projections, and experts in the key and query projections were found to be unnecessary*. Any proposal

that routes the key projection must engage with this result. PHASOR does not route the key *projection*: it uses the router to determine the *sparsity pattern* (which keys a query may see), which is an orthogonal axis to which weight matrix produces the key. We consider this an important distinction and one that the SwitchHead result sharpens rather than contradicts.

2.4 Conditional computation over tokens

Mixture-of-Depths [32] routes a capped number of tokens through each block; the rest take the residual path. This is the mechanism our fallback modifies. Related inference-time methods select which cached keys to read per query [38, 45]; they are complementary, as they do not change training.

2.5 Rotary embeddings

RoPE [37] multiplies each 2-D subvector of $\mathbf{q}$ and $\mathbf{k}$ by a rotation whose angle is $\theta_j m$ at position m, so that $\langle \mathbf{q}_m, \mathbf{k}_n \rangle$ depends on $m-n$. Applying RoPE to only a fraction of head dimensions is common [6], and DeepSeek-V2 [16] explicitly carries *decoupled* RoPE dimensions alongside compressed latent dimensions. Proposition 2 gives a routing-specific reason for the same structural choice. The nearest analysis we are aware of is Chiang and Yogatama [9], who argue that RoPE causes dimension inefficiency in attention heads used for long-distance retrieval; that argument concerns the *score*, ours concerns the *partition*, but both point at the same underlying fact that rotary phase consumes angular capacity that content would otherwise use.

3 Method

3.1 Notation and head decomposition

Let $X \in \mathbb{R}^{n \times d}$ be the layer input for a sequence of length n. For a single head with dimension d_h we compute $\mathbf{q}_i, \mathbf{k}_i \in \mathbb{R}^{d_h}$ and $\mathbf{v}_i \in \mathbb{R}^{d_v}$ as usual. We split the head dimension into a positional part and a content part,

$$\mathbf{q}_i = [\,\mathbf{q}_i^{\mathrm{p}} \,;\, \mathbf{q}_i^{\mathrm{c}}\,], \qquad \mathbf{k}_i = [\,\mathbf{k}_i^{\mathrm{p}} \,;\, \mathbf{k}_i^{\mathrm{c}}\,], \qquad \mathbf{q}_i^{\mathrm{p}}, \mathbf{k}_i^{\mathrm{p}} \in \mathbb{R}^{d_{\mathrm{p}}},\ \ \mathbf{q}_i^{\mathrm{c}}, \mathbf{k}_i^{\mathrm{c}} \in \mathbb{R}^{d_{\mathrm{c}}}, \tag{1}$$

with $d_{\mathrm{p}} + d_{\mathrm{c}} = d_h$. RoPE is applied to the positional part only:

$$\tilde{\mathbf{q}}_i = [\,\Theta_i \mathbf{q}_i^{\mathrm{p}} \,;\, \mathbf{q}_i^{\mathrm{c}}\,], \qquad \tilde{\mathbf{k}}_i = [\,\Theta_i \mathbf{k}_i^{\mathrm{p}} \,;\, \mathbf{k}_i^{\mathrm{c}}\,], \tag{2}$$

where Θ_i is the block-diagonal RoPE matrix at position i. Attention scores use the full $\tilde{\mathbf{q}}, \tilde{\mathbf{k}}$, so positional information is not removed from the model; it is removed only from the *router's view*. Setting $d_{\mathrm{p}} = d_h$ recovers standard full RoPE and, by Proposition 2, destroys content routing; setting $d_{\mathrm{p}} = 0$ recovers a NoPE model. We treat $\rho = d_{\mathrm{p}}/d_h$ as a hyperparameter and ablate it.

3.2 The orthogonal rotational router

Definition 1 (PHASOR router)**.** *Let $\mathbf{R} \in O(d_{\mathrm{c}})$ be a learned orthogonal matrix and let $C = \{\pm\mathbf{e}_1, \ldots, \pm\mathbf{e}_{d_{\mathrm{c}}}\}$ be the fixed cross-polytope codebook, so $B = 2d_{\mathrm{c}}$. For a routing input $\mathbf{u} \in \mathbb{R}^{d_{\mathrm{c}}}$ define*

$$\hat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|_2 + \varepsilon}, \qquad \mathbf{s}(\mathbf{u}) = \big[\,\mathbf{R}\hat{\mathbf{u}} \,;\, -\mathbf{R}\hat{\mathbf{u}}\,\big] \in \mathbb{R}^B, \qquad b(\mathbf{u}) = \arg\max_{j \in [B]} s_j(\mathbf{u}). \tag{3}$$

Queries and keys are routed by the same map applied to their content parts: $b_i^q = b(\mathbf{q}_i^{\mathrm{c}})$ and $b_i^k = b(\mathbf{k}_i^{\mathrm{c}})$. Unlike Reformer we do not tie $\mathbf{q} = \mathbf{k}$; we tie the *router*, which is sufficient for queries and keys of the same type to land in the same sector while leaving the attention bilinear form unconstrained.

Parametrisation of R. We use the Cayley transform, $\mathbf{R} = (\mathbf{I} - A)(\mathbf{I} + A)^{-1}$ with $A = -A^\top$ skew-symmetric, which covers the connected component of $O(d_c)$ containing the identity and is differentiable everywhere it is defined. Two alternatives with the same guarantee are a product of p Householder reflections [29], giving $O(pd_c)$ cost and a tunable expressivity/cost trade-off, and the matrix exponential with trivialisation [3, 25]. Since d_c is a head content dimension (typically 32–64), the $O(d_c^3)$ cost of the Cayley inverse is paid once per layer per step, not per token, and is negligible; we still report Householder as the default for large d_c.

Why not a free codebook. With learned prototypes $\{\mathbf{c}_b\}$ the rotation is absorbed: $\langle \mathbf{R}\hat{\mathbf{u}}, \mathbf{c}_b \rangle = \langle \hat{\mathbf{u}}, \mathbf{R}^\top \mathbf{c}_b \rangle$. Definition 1 is therefore *not* a reparametrisation of a cosine router; it is a strictly constrained subfamily, and the constraint has a consequence:

Proposition 1 (Codebook non-collapse)**.** *Let $\mathbf{p}_b = \mathbf{R}^\top \mathbf{c}_b$ be the effective prototype of bucket b under Definition 1. Then for every $\mathbf{R} \in O(d_c)$ and every $b \neq b'$ with $\mathbf{p}_{b'} \neq -\mathbf{p}_b$,*

$$\angle(\mathbf{p}_b, \mathbf{p}_{b'}) = \frac{\pi}{2}, \tag{4}$$

and $\|\mathbf{p}_b\| = 1$ for all b. In particular $\min_{b \neq b'} \angle(\mathbf{p}_b, \mathbf{p}_{b'})$ is invariant to the parameters. By contrast, for an unconstrained linear router with prototypes $\mathbf{p}_b = \mathbf{W}^\top \mathbf{e}_b$ the same quantity can be driven to 0 by gradient descent.

Proof. $\mathbf{R}$ orthogonal preserves inner products, so $\langle \mathbf{p}_b, \mathbf{p}_{b'} \rangle = \langle \mathbf{R}^\top \mathbf{c}_b, \mathbf{R}^\top \mathbf{c}_{b'} \rangle = \langle \mathbf{c}_b, \mathbf{c}_{b'} \rangle$. For the cross-polytope codebook, $\langle \mathbf{c}_b, \mathbf{c}_{b'} \rangle \in \{1, 0, -1\}$, taking value 0 for all non-identical, non-antipodal pairs, and $\|\mathbf{c}_b\| = 1$. The claim for the unconstrained case follows because $\mathbf{W}$ ranges over all of $\mathbb{R}^{d_c \times B}$, which includes rank-one matrices. □

Proposition 1 is a statement about the router's *prototypes*, and we want to be exact about its scope. It says that the partition of the sphere induced by the router remains a rigid, equiangular cross-polytope partition throughout training, so the failure mode identified by Chi et al. [8] — prototypes drifting together until the router cannot discriminate — cannot occur in this family. It says nothing about the *distribution of data* over that partition. If the encoder learns to map most tokens into one sector, load collapses even though the partition does not. Section 3.5 treats that separately, and Section 4.4 lists it as the main open risk.

Remark 1 (Relation to Reformer and Routing Transformer)**.** *Fixing $\mathbf{R}$ to a random draw and freezing it recovers Reformer's hash function exactly. Replacing the fixed codebook by EMA-updated free centroids and dropping orthogonality recovers Routing Transformer's assignment rule up to the optimiser. PHASOR is the diagonal of that square: learned frame, fixed codebook.*

3.3 Phase contamination: why the router must not read RoPE dimensions

Consider a router that reads a subspace to which RoPE has been applied. Restrict attention to one RoPE plane with frequency θ and let $u \in \mathbb{R}^2$ be the content vector in that plane, $\alpha = \arg(u)$. At position m the routed vector is $\Theta_m^{(\theta)} u$ with argument $\alpha + \theta m$. Any angular router partitions this plane into B_2 arcs.

Proposition 2 (Phase contamination)**.** *Let two tokens have identical content in the plane ($\alpha_1 = \alpha_2 = \alpha$) and occupy positions m and n, and let the router assign buckets by the arc of width $2\pi/B_2$ containing the argument. Write $\Delta = m - n$. Then the two tokens are co-routed if and only if*

$$\left\lfloor \frac{B_2(\alpha + \theta m)}{2\pi} \right\rfloor \equiv \left\lfloor \frac{B_2(\alpha + \theta n)}{2\pi} \right\rfloor \pmod{B_2}. \tag{5}$$

If $\theta/2\pi$ is irrational, then by Weyl equidistribution the sequence $(\theta\Delta \bmod 2\pi)_{\Delta\in\mathbb{N}}$ is equidistributed on $[0,2\pi)$, and for Δ drawn uniformly from $\{1,\dots,W\}$,

$$\lim_{W\to\infty} \mathbb{P}\big[\textit{co-routed}\big] = \frac{1}{B_2}, \tag{6}$$

i.e. chance level: the content agreement $\alpha_1 = \alpha_2$ confers no advantage. More generally the co-routing probability for a window of width W is $\frac{1}{B_2} + O\big(\frac{1}{\theta W}\big)$, so contamination is severe for high-frequency planes ($\theta W \gg 2\pi$) and mild for low-frequency ones ($\theta W \ll 2\pi$).

Proof sketch. The arc index is a function of $\alpha + \theta m \bmod 2\pi$ alone. Two indices coincide iff the phases fall in the same arc; since both phases share the offset α, the event depends only on $\theta\Delta \bmod 2\pi$ and on the position of α within its arc. Averaging over the equidistributed $\theta\Delta$ gives arc-length $2\pi/B_2$ over circumference 2π. The finite-window remainder follows from the discrepancy of $\{\theta\Delta\}$ over W terms. □

The practical reading is that *a clustering-based sparse attention layer stacked naively on RoPE does not cluster by content*. Its high-frequency planes hash by position; its low-frequency planes retain content structure but contribute little angular resolution. Since RoPE frequencies are geometrically spaced over several orders of magnitude, a router reading all dimensions gets a mixture in which the position-hashing component dominates for typical context lengths. This also suggests a diagnostic that does not require training a new model, which we include as experiment E0 in Section 6: if it fails to show contamination, the motivation for our design weakens considerably.

Remark 2 (The aggregate limit is not $1/B$)**.** *Proposition 2 concerns a single plane, and the extrapolation to a whole head does not follow from it. Measurement (Section 5.2) shows that co-routing agreement under RoPE-read routing plateaus one to two orders of magnitude above $1/B$, because the router reaches only five to ten* effective *buckets however many are provisioned. Proposition 3 gives the reason: varying position alone confines the routing input to a torus of dimension $d_c/2$, a measure-zero subset of the sphere, so most of the partition is unreachable. The correct statement of the aggregate effect is therefore not that routing degenerates to uniform hashing but that it degenerates to hashing over a small effective codebook — a worse failure, since the model loses content selectivity* and *bucket capacity, and the relative loss grows with B.*

PHASOR routes on $\mathbf{q}^c, \mathbf{k}^c$, which by construction are not rotated by Θ_i. Bucket identity is then a function of content alone, exactly, and positional information reaches the score through $\mathbf{q}^p, \mathbf{k}^p$ as usual. We regard this decoupling as the most defensible of our three contributions, in the sense that it is the one that follows from an argument rather than from a hope.

3.4 Sparse pattern and the attention-visible residue

Let $\mathcal{K}_b = \{j \le i : b_j^k = b\}$ be the set of causally admissible keys in bucket b. The naive pattern lets query i attend to $\mathcal{K}_{b_i^q}$. We augment it as follows.

Residue summaries. For every bucket b define a pooled summary of its keys and values,

$$\bar{\mathbf{k}}_b = \frac{1}{|\mathcal{K}_b|}\sum_{j\in\mathcal{K}_b} \mathbf{k}_j, \qquad \bar{\mathbf{v}}_b = \frac{1}{|\mathcal{K}_b|}\sum_{j\in\mathcal{K}_b} \mathbf{v}_j, \tag{7}$$

computed causally by running means (so they are available at decoding time at $O(B)$ state). The key set of query i is then

$$\mathcal{S}_i = \underbrace{\mathcal{K}_{b_i^q}}_{\text{fine, in-bucket}} \cup \underbrace{\{\bar{\mathbf{k}}_b : b \neq b_i^q\}}_{\text{coarse, residue}} \cup \underbrace{\mathcal{W}_i}_{\text{local window}}, \tag{8}$$

where $\mathscr{W}_i$ is a short sliding window of w recent tokens, included because local dependencies are dense and should not depend on routing luck (the same reasoning as NSA's window branch). The output is a single softmax over $\mathscr{S}_i$ — not a mixture of separate attentions — so the model allocates mass between fine and coarse evidence within one normalisation.

Why this and not a skip. In MoD-style routing an unselected token contributes nothing to the block and is carried by the residual stream. Information is preserved across depth, but within the layer the model cannot distinguish "considered and found irrelevant" from "not considered". Making the residue visible at coarse resolution restores that distinction: the softmax can put mass on $\bar{\mathbf{v}}_b$ when the excluded material matters, and the gradient of that mass flows back into the router through $\bar{\mathbf{k}}_b$, giving the router a corrective signal it does not otherwise receive. This is the mechanism by which the "it would be filtered by attention anyway" intuition becomes true rather than assumed; under a pure skip it is false.

Cost. B extra keys per query, hence $O(nB)$ additional score computations and $O(Bd_h)$ additional cache state, both linear in n with a small constant ($B = 2d_c \approx 64$–128).

Residual path. We retain the residual connection unchanged, so PHASOR is a strict superset of the MoD fallback rather than a replacement for it.

3.5 Differentiability and training

The assignment $b(\cdot)$ is an arg max and is not differentiable. We use three standard devices and report their ablation rather than claiming one is right.

1. **Score gating.** As in Switch [17] and MoD [32], each attended key carries its router score as a multiplicative gate, $\tilde{s}_i = \sigma(s_{b_i^q}(\mathbf{q}_i^c))$, applied to the attention output. Gradients then reach $\mathbf{R}$ through the magnitude of the winning score even though the index is discrete.
2. **Straight-through relaxation.** A temperature-τ softmax over $\mathbf{s}$ in the backward pass with hard assignment in the forward pass [5, 19, 27], annealed over training.
3. **Residue gradient.** As noted in Section 3.4, attention mass on $\bar{\mathbf{k}}_b$ provides gradient to keys the router excluded, which a pure skip does not.

Balancing. Proposition 1 does not bound the load. We therefore keep an auxiliary loss. Because the partition is rigid, the natural form is an entropy term on the marginal bucket distribution $\pi_b = \frac{1}{n}\sum_i \mathbb{1}[b_i^k = b]$,

$$\mathscr{L}_{\text{bal}} = -\lambda\, H(\pi) = \lambda \sum_{b=1}^{B} \pi_b \log \pi_b, \tag{9}$$

optionally combined with the standard Switch load-balancing product form. Our hypothesis — and it is a hypothesis, to be tested in ablation A6 — is that the required λ is *smaller* than for an unconstrained router, because one of the two collapse channels has been closed by construction. We do not claim the loss can be removed. Gemini-style claims that orthogonality removes the need for balancing do not follow from Proposition 1 and we do not make them.

Multi-round routing. Following Reformer's multi-round hashing, we may use $n_r > 1$ rotations $\mathbf{R}^{(1)}, \ldots, \mathbf{R}^{(n_r)}$ and take the union of the induced key sets, reducing the chance that a relevant key is separated by an unlucky boundary. Cost scales linearly in n_r.

3.6 Design decisions the method statement does not determine

Writing the reference implementation forced five choices that the description above leaves open. We list them because a reader cannot reconstruct them from the method, because each is a plausible place for the design to be wrong, and because at least two of them plausibly matter as much as the contributions.

D1. Which space the residue summaries live in. Section 3.4 writes $\bar{\mathbf{k}}_b$ as a mean of keys without saying whether the keys are pre- or post-RoPE. It matters: averaging post-RoPE keys averages vectors at different phases, so the mean suffers destructive interference and shrinks as the bucket grows — the summary degrades exactly when it carries the most tokens. We pool the *content* part of keys and score them against $\mathbf{q}^{\mathrm{c}}$; values are pooled whole, since they carry no positional phase. The consequence is that the residue branch has no positional information at all. That is defensible for a coarse global branch, in the spirit of registers [15] and sinks [41], but it is a decision and not a detail.

D2. Calibrating two branches inside one softmax. Fine logits are $\mathbf{q}^{\top}\mathbf{k}$ over d_h dimensions scaled by $1/\sqrt{d_h}$; residue logits are $\mathbf{q}^{\mathrm{c}\top}\bar{\mathbf{k}}^{\mathrm{c}}$ over d_{c} dimensions. Placing both in one softmax at the same scale is not obviously right, so the residue branch carries a learnable per-head scale and bias. The bias initialisation turns out to control the branch's share of attention mass over nearly two orders of magnitude (Section 5.6), which makes it a hyperparameter the method must declare rather than a detail it can omit.

D3. Which round's buckets define the summaries. With $n_r > 1$ a key belongs to a different bucket in each round. Maintaining summaries per round multiplies the state by n_r; we use round 0 only. This is arbitrary and we have not measured the alternative.

D4. The window and the summaries overlap. A key inside the sliding window that belongs to a non-selected bucket is attended finely *and* contributes to that bucket's summary. The effect is small, but it means the residue branch is not exactly "the complement of what the router selected" — which is the phrase we use to distinguish it from NSA's compression branch. Either the phrase or the construction should be tightened.

D5. The balancing loss reads future tokens. $\mathscr{L}_{\mathrm{bal}}$ is computed over all positions in the batch, including positions after the current one. This does not break autoregression: it is a batch-level regulariser on parameters, exactly as in standard MoE auxiliary losses, and the model *function* remains causal. We state it because it is the kind of thing a reader should not have to verify for themselves, and because the released implementation tests the causality of the model function by autograd rather than by inspection.

3.7 The layer

Algorithm 1 gives the full layer. Figure 1 illustrates the routing geometry and the RoPE decoupling.

Algorithm 1 PHASOR attention (single head, causal)

Require: $X \in \mathbb{R}^{n\times d}$; projections $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$; rotation parameters A; split $(d_\mathrm{p}, d_\mathrm{c})$; window w; rounds n_r

$Q, K, V \leftarrow X\mathbf{W}_Q,\ X\mathbf{W}_K,\ X\mathbf{W}_V$

split $Q \to (Q^\mathrm{p}, Q^\mathrm{c})$, $K \to (K^\mathrm{p}, K^\mathrm{c})$ ▷ $d_\mathrm{p} + d_\mathrm{c} = d_h$

$\tilde{Q} \leftarrow [\mathrm{RoPE}(Q^\mathrm{p}); Q^\mathrm{c}]$; $\quad \tilde{K} \leftarrow [\mathrm{RoPE}(K^\mathrm{p}); K^\mathrm{c}]$

$\mathbf{R} \leftarrow \mathrm{Cayley}(A)$ ▷ or $\prod_{t\le p}(\mathbf{I} - 2\mathbf{h}_t\mathbf{h}_t^\top)$

for $r = 1$ to n_r **do**

  $b^q \leftarrow \arg\max\left[\mathbf{R}^{(r)}\widehat{Q^\mathrm{c}}; -\mathbf{R}^{(r)}\widehat{Q^\mathrm{c}}\right]$; $\quad b^k \leftarrow \arg\max\left[\mathbf{R}^{(r)}\widehat{K^\mathrm{c}}; -\mathbf{R}^{(r)}\widehat{K^\mathrm{c}}\right]$

  sort tokens by $(b^k, \text{position})$ and form equal-size chunks ▷ as in Reformer

end for

$\bar{K}, \bar{V} \leftarrow$ causal running means of K, V within each bucket ▷ B entries

for each query i **do**

  $\mathscr{S}_i \leftarrow \mathscr{K}_{b_i^q} \cup \{\bar{\mathbf{k}}_b\}_{b\neq b_i^q} \cup \mathscr{W}_i$ ▷ union over rounds

  $O_i \leftarrow \mathrm{softmax}\big(\tilde{\mathbf{q}}_i^\top \tilde{\mathbf{k}}_j/\sqrt{d_h}\big)_{j\in\mathscr{S}_i} \cdot V_{\mathscr{S}_i}$

  $O_i \leftarrow \sigma\big(s_{b_i^q}\big) \cdot O_i$ ▷ score gate

end for

return $O\mathbf{W}_O$; $\quad$ add $\mathscr{L}_\mathrm{bal}$ to the training objective

4 Analysis

4.1 Complexity, and the condition it depends on

Let n be the sequence length, $B = 2d_\mathrm{c}$ the number of buckets, n_r the number of rounds and w the window. With chunked bucketing at chunk size c, per head and per layer:

$$\underbrace{O\big(n_r\, n\, c\big)}_{\text{in-bucket}} + \underbrace{O(nB)}_{\text{residue}} + \underbrace{O(nw)}_{\text{window}} + \underbrace{O(n\, d_\mathrm{c}\log n)}_{\text{routing + sort}}. \tag{10}$$

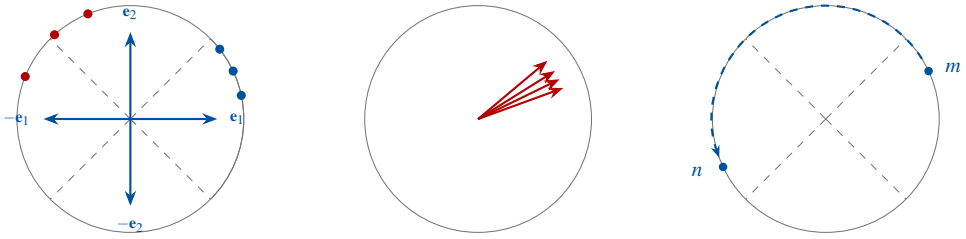
Taking $c \approx n/B$ and balancing the first two terms gives $B \asymp \sqrt{n_r}\sqrt{n}$ and total cost $O(\sqrt{n_r}\, n^{1.5})$, the same asymptotic regime as Routing Transformer.

That derivation hides a condition, and measuring the implementation brought it out (Section 5.7). The chunk size cannot be chosen freely: fixed-size chunking approximates bucket-wise attention faithfully only when c is comparable to the mean bucket size $\bar{m} = n/B$, and at $c = \bar{m}/4$ the chunked pattern retains 38% of true in-bucket pairs. So $c \gtrsim n/B$ is a correctness constraint, not a tuning choice, and the in-bucket term is $\Omega(n^2/B)$. The $O(n^{1.5})$ figure therefore holds only when

$$B \asymp \sqrt{n}, \qquad \text{equivalently} \qquad d_\mathrm{c} \approx \tfrac{1}{2}\sqrt{n}, \tag{11}$$

since Definition 1 ties B to twice the routing width. At fixed d_c the method is quadratic with constant $1/B$. We state this as a property of the method rather than of the implementation, because it follows from the definition of the router and not from any coding choice.

Remark 3 (The width–context coupling)**.** *Reaching the claimed operating point at $n = 4096$ needs $d_\mathrm{c} = 32$, which is ordinary; at $n = 65{,}536$ it needs $d_\mathrm{c} = 128$, a content subspace larger than most whole heads. Multi-round routing is the escape hatch — n_r independent partitions unioned together behave like a larger effective codebook at cost linear in n_r — but whether unioning rounds recovers the fidelity a single wide partition would have is an open question we have not measured. This coupling is, in our view, the most consequential limitation of the design as specified.*



(a) fixed codebook, learned **R**
sectors always $\pi/2$ apart

(b) free prototypes
can collapse together

(c) same content, RoPE'd
phase $\theta\Delta$ moves the bucket

Figure 1: Routing geometry. (a) A learned orthogonal **R** acting on a fixed cross-polytope codebook: the partition is rigid, sector prototypes stay $\pi/2$ apart for every parameter value (Proposition 1). (b) An unconstrained router's prototypes may concentrate — representation collapse [8]. (c) If routing reads RoPE-rotated dimensions, two tokens with *identical content* at positions m and n are separated by phase $\theta(m-n)$ and land in different sectors, so content routing degrades toward position hashing (Proposition 2).

4.2 What is *not* reduced: KV cache

We state this explicitly because it is a common overclaim. Routing changes *which* keys a query reads; it does not change how many keys exist. Every token still produces exactly one $(\mathbf{k}, \mathbf{v})$ pair, so the KV cache remains $\Theta(nd_h)$ per head, plus $O(Bd_h)$ for the running summaries. PHASOR reduces read bandwidth per decoding step (from n to roughly $n/B + B + w$), which is the quantity that dominates memory-bound decoding, but it does not reduce cache *capacity*. Reducing capacity requires a different axis — MQA/GQA [1, 35], latent compression [16], or eviction [45]. These compose with PHASOR and are orthogonal to it.

Similarly, routing *which projection matrix* produces a key (the MoE-in-attention axis of MoA, SwitchHead, MoH) does not by itself reduce the quadratic term at all: each token still emits a key and each query still scores against it. Only the sparsity *pattern* reduces the quadratic term. PHASOR acts on the pattern.

4.3 Norm preservation

Since **R** is orthogonal and $\hat{\mathbf{u}}$ is unit norm, $\|\mathbf{R}\hat{\mathbf{u}}\| = 1$ and $\max_j s_j \in [1/\sqrt{d_c}, 1]$, with the lower bound attained at the maximally ambiguous point (equal projection onto all axes) and the upper bound when $\hat{\mathbf{u}}$ aligns with a codebook direction. The winning score is therefore a bounded, well-scaled confidence signal, which is convenient for the gate $\sigma(s)$ and removes the need for the router z-loss used to control logit magnitude in unconstrained routers [46]. We flag this as a *plausible* secondary benefit rather than a demonstrated one.

4.4 Failure modes we expect

1. **Load collapse survives, and it is entirely the encoder's doing.** Proposition 1 closes the codebook channel only. Implementing the router sharpened this in both directions. If the encoder concentrates content directions, buckets become unbalanced and the n^2/B argument fails. This is the single most likely way the method underperforms.

   Section 5.3 measures this and Proposition 4 explains it: the load is a function of the routing input's anisotropy and is invariant to the router's parameters in the isotropic case. The practical reading is that a balancing term has to act on the encoder, not on **R**.

2. **Rigidity may cost accuracy.** A fixed equiangular partition is less expressive than data-fit centroids.

Routing Transformer's k-means can allocate resolution where the data is dense; PHASOR cannot, except through $\mathbf{R}$, which moves the whole frame rigidly. If the data manifold is anisotropic, this is a real disadvantage. A2 measures it.

3. **Semantic specialisation may not appear.** The intuition motivating this line of work is that buckets will correspond to linguistically meaningful categories. Evidence from MoE models suggests expert specialisation tends to be syntactic and positional rather than semantic [20], and probing work shows linguistic structure is distributed across layers rather than concentrated [12, 39]. We therefore do *not* rest any efficiency claim on interpretability of buckets; we measure it (A7) as a descriptive question, and a negative result there does not falsify the method.

4. **The fixed-hash null.** Roller et al. [33] show random fixed hashing is a strong router. If A1 (learned vs. random $\mathbf{R}$) shows no gap, the central claim of this paper is false and should be reported as such.

5. **Hardware.** Gather/scatter over sorted buckets has poor arithmetic intensity relative to dense FlashAttention [14]. Reformer and Routing Transformer both report that theoretical FLOP savings translate imperfectly into wall-clock. NSA [42] exists precisely because of this. We therefore require wall-clock, not only FLOPs, in every table.

5 Controlled Numerical Studies

5.1 What this evidence is, and what it cannot support

Everything in this section is a measurement of the *mechanism*, taken against the reference implementation on constructed inputs whose ground truth we control. None of it is a language-modelling result, and none of it can substitute for one.

The reason to run these studies before training anything is that they are exact, cheap, and capable of falsifying claims. A perplexity number cannot tell you whether the router's partition collapsed, whether the load imbalance came from the router or the encoder, or whether the fast kernel computes the function the method describes; it aggregates all of those into one scalar that is also sensitive to a dozen unrelated choices. The studies below separate them. Two of them returned answers that contradicted what this paper previously asserted, which is the main argument for doing them in this order.

What they cannot show: whether any of this helps a trained model. In particular, every study here uses an *untrained* router, so they measure the geometry the architecture imposes rather than what optimisation could make of it. Section 5.5 is the sharpest illustration — it finds no difference between ablation arms at initialisation at all, which is itself a constraint on how the remaining claims can be tested.

Reproduction. Each study writes a CSV; the tables and figures in this section are generated from those CSVs by a script, so no number here is transcribed by hand and the evidence regenerates end to end:

```
python experiments/numerical_studies.py && python experiments/make_tables.py
```

Unless stated otherwise: d_c and the bucket count $B = 2d_c$ as given per row; RoPE base 10^4; Cayley-parametrised rotations at default initialisation; results averaged over 32 content directions for the routing studies and 4–8 random initialisations elsewhere; CPU, float32, seeds fixed and recorded.

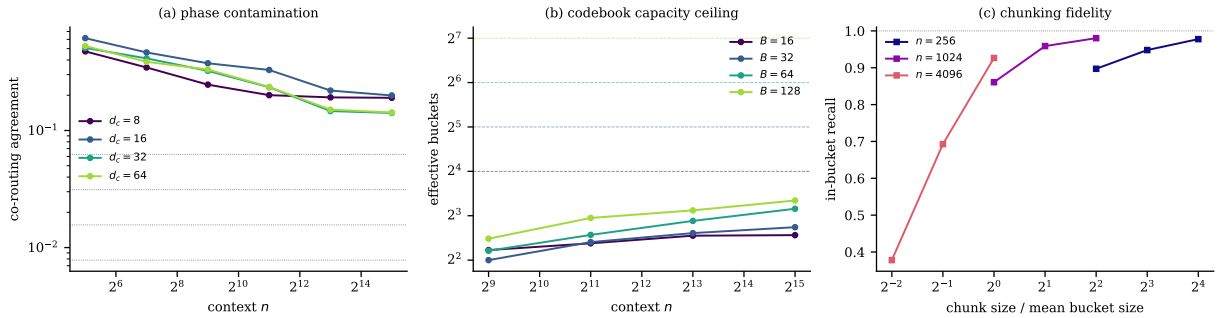


Figure 2: The three findings that changed the paper. **(a)** Co-routing agreement for *content-identical* tokens whose only difference is position, under RoPE-read routing, against context length. Dotted lines mark chance $1/B$ for each width. Agreement falls monotonically, as Proposition 2 predicts, but plateaus an order of magnitude above chance. **(b)** The reason: the number of *effective* buckets the router reaches saturates near 6–10 (dashed lines mark the provisioned B). Provisioning more buckets does not raise the ceiling, so the relative collapse worsens with B. **(c)** Fidelity of bucket-sorted fixed-size chunking to true bucket-wise attention, against the ratio of chunk size to mean bucket size. The three context lengths collapse onto one curve, identifying $c/\bar{m}$ as the governing variable; fidelity is lost below $c \approx \bar{m}$.

5.2 S1 and S2: phase contamination and the capacity ceiling

Construction. A content vector $\mathbf{u}$ is drawn and held fixed. It is placed at every position $0, \ldots, n-1$, RoPE is applied, and the router is read. Any loss of co-routing is attributable to rotary phase and to nothing else, because the content never varies. We report the fraction of positions whose bucket matches position 0 ("co-routing agreement"), and the perplexity of the visited-bucket distribution ("effective buckets"), averaged over 32 content directions.

		co-routing agreement				effective buckets			
d_c	B	512	2048	8192	32768	512	2048	8192	32768
8	16	0.2462	0.2003	0.1917	0.1901	4.68	5.2	5.86	5.91
16	32	0.375	0.3287	0.2193	0.1991	4.0	5.3	6.1	6.69
32	64	0.3229	0.2341	0.1465	0.1411	4.63	5.93	7.38	8.92
64	128	0.3314	0.2356	0.1506	0.1424	5.59	7.73	8.7	10.15

Table 1: Phase contamination (S1) and codebook capacity (S2), by routing width and context length. Chance agreement is $1/B$: 0.063, 0.031, 0.016, 0.008 for the four widths. Agreement plateaus at roughly 0.14–0.20 for every width, so its *ratio* to chance grows from 3× at $B = 16$ to 18× at $B = 128$. The effective bucket count explains this: it saturates near 6–10 regardless of B.

What we had claimed. Proposition 2 shows that for a *single* rotary plane the co-routing probability tends to $1/B_2$ under equidistribution. We extrapolated from that to the whole head and wrote that content routing "degenerates into position hashing".

What the measurement shows. The monotone decay is there and is robust: agreement falls from 0.61 to 0.20 at $d_c = 16$, and the fall continues across four orders of magnitude of context. But it plateaus, and the plateau sits at 3 to 18 times chance depending on width. The effective bucket count in the right half of

Table 1 identifies the cause: the router never reaches more than about ten buckets, however many exist. At $B = 16$ that is 37% of the codebook; at $B = 128$ it is 8%.

Correction to the paper's claim

Routing on rotary dimensions does not degrade toward uniform hashing. It degrades toward a **capacity ceiling**: a small, nearly width-independent number of reachable buckets. This is a worse failure than uniform hashing, not a milder one — the model loses content selectivity *and* most of its bucket capacity, and the loss of capacity gets relatively worse the more capacity is provisioned. Proposition 2 stands as stated, per plane. Its extrapolation to the head does not, and we have replaced it.

Why. The mechanism is geometric and exact.

Proposition 3 (Torus confinement)**.** *Let $\hat{\mathbf{u}} \in S^{d_c-1}$ be a fixed unit content vector with $d_c = 2h$, and let $\mathbf{w}(m) = \Theta_m \hat{\mathbf{u}}$ be its RoPE image at position m, where Θ_m is block-diagonal with h planar rotations of angles $\theta_1 m, \ldots, \theta_h m$. Then for all m,*

$$\mathbf{w}(m) \in \mathscr{T}(\hat{\mathbf{u}}) \;=\; \Big\{ \mathbf{x} \in \mathbb{R}^{d_c} : \|(x_j, x_{j+h})\| = E_j(\hat{\mathbf{u}}) \;\; \forall j \le h \Big\}, \tag{12}$$

an h-dimensional torus embedded in S^{d_c-1}, where E_j is the fixed energy of plane j. Consequently the set of routing inputs reachable by varying position alone has measure zero in S^{d_c-1}, and the set of buckets reachable by $\hat{\mathbf{u}}$ is bounded by the number of cells of the partition that $\mathbf{R}\,\mathscr{T}(\hat{\mathbf{u}})$ meets — a quantity governed by $h = d_c/2$, not by B.

Proof. Each 2×2 block of Θ_m is a rotation and therefore preserves the norm of the corresponding coordinate pair, so E_j is constant in m. The set of points with all h plane norms fixed is a product of h circles. Its dimension is $h = d_c/2 < d_c - 1$ for $d_c > 2$, hence measure zero in the sphere. The reachable bucket set is the image of this torus under the orthogonal $\mathbf{R}$, intersected with the B cells of the cross-polytope partition; an h-dimensional set meets a number of cells that grows with h, and $\mathbf{R}$ being orthogonal does not change the dimension. □

This predicts slow growth of the ceiling with d_c and no dependence on B beyond the $B = 2d_c$ coupling, which is what Table 1 shows: quadrupling h from 4 to 32 moves the effective bucket count only from 5.9 to 10.2. It also explains why the plateau value of agreement is roughly width-independent while its ratio to chance is not.

Consequence for the protocol. The diagnostic E0 must report *effective bucket count*, not only mutual information between bucket and position. We also found that mutual information with binned position is non-monotone in context length — it peaks when the phase has advanced enough to correlate with position but not enough to scramble it — and is therefore a misleading headline metric. The quantity that falls monotonically is the mutual information between cluster and *token identity*. Section 6 reflects both corrections.

5.3 S3: the load belongs to the data, not to the router

Construction. Routing inputs are drawn as $\mathbf{u} \sim \mathcal{N}(\kappa \mathbf{e}_1, \mathbf{I})$ for a concentration parameter κ interpolating from isotropic ($\kappa = 0$) to strongly anisotropic. For each, we record the normalised load entropy at initialisation, and again after 300 Adam steps on an objective that explicitly rewards pushing every token into bucket 0 — the most direct collapse pressure available to the router.

κ	$H(\pi)/\log B$ at init	after pressure	Δ	buckets used
0.0	0.9994	0.9994	-0.0	32
0.25	0.9968	0.9967	-0.0001	32
0.5	0.9877	0.9884	0.0007	32
1.0	0.9473	0.9476	0.0003	32
2.0	0.7151	0.7059	-0.0092	32
4.0	0.1024	0.0787	-0.0237	31

Table 2: Load entropy against input anisotropy κ, before and after 300 steps of explicit collapse pressure on the rotation ($d_c = 16$, $B = 32$, $n = 8192$). The load is determined almost entirely by κ; the pressure moves it by at most 0.024.

What the measurement shows. Load entropy is a clean monotone function of input concentration, falling from 0.999 at $\kappa = 0$ to 0.102 at $\kappa = 4$. The effect of 300 steps of adversarial optimisation on the router is at most 0.024 and is usually within noise of zero. At the isotropic end it is exactly zero to four decimals. This is not an artefact of the optimiser; it is forced:

Proposition 4 (Load is a property of the data)**.** *Let the routing input* $\mathbf{u}$ *be drawn from a rotationally invariant distribution on* $\mathbb{R}^{d_c}$*. Then under Definition 1 the bucket marginal* π *is uniform for* every $\mathbf{R} \in O(d_c)$*, and is therefore invariant to the router's parameters.*

Proof. If $\mathbf{u}$ is isotropic then so is $\mathbf{R}\hat{\mathbf{u}}$ for orthogonal $\mathbf{R}$, since orthogonal maps preserve rotational invariance. The cross-polytope sectors are congruent under the symmetry group of the codebook, so an isotropic direction falls in each with equal probability. □

This invalidated one of our own ablations

Ablation A6 was specified as a sweep over the coefficient λ of a balancing loss placed on the router, to test whether orthogonality reduces the balancing needed. Proposition 4 says the experiment cannot work: a loss whose gradient reaches only $\mathbf{R}$ cannot change a marginal that does not depend on $\mathbf{R}$. Balancing has to act on the *encoder* that produces the routing input, or the input must be whitened before the router sees it. A6 is respecified in Section 6 with a whitening arm, and we flag that the original design would have produced a null result that we might have misread as evidence for the method.

Two further consequences. First, Section 4.4's "load collapse survives" is now a sharper and more testable claim: load collapse is attributable *entirely* to anisotropy of the encoder's output. Second, Proposition 4 is a genuine guarantee in the isotropic regime and comes for free with the orthogonality constraint, which no unconstrained router can offer.

5.4 S4: the codebook geometry is rigid, and a free router's is not

Construction. Proposition 1 is a statement about reachability: the minimum prototype angle of an unconstrained router *can* be driven to zero by gradient descent, and ours cannot, for any parameter value. The honest test is therefore adversarial rather than task-shaped — we optimise directly against the quantity in both families, maximising mean pairwise prototype cosine similarity, and watch the minimum angle.

What the measurement shows. Exactly what Proposition 1 asserts, with no slack: the orthogonal router's minimum prototype angle is 90.000° at step 0 and 90.000° at step 400, while the unconstrained router goes

router	step 0	step 50	step 150	step 400
orthogonal (ours)	90.0	90.0	90.0	90.0
unconstrained linear	48.43	0.809	0.0	0.0

Table 3: Minimum angle between distinct, non-antipodal bucket prototypes (degrees) under an objective that explicitly rewards collapse ($d_c = 16$, $B = 32$). The orthogonal router is unmoved at every step; the unconstrained router collapses completely within 150 steps.

from 48.4° to 0.8° in 50 steps and to 0.0° by 150. The orthogonality is exact rather than penalty-enforced, so this is a structural property and not a tuning outcome.

What it does not show. That this matters for language modelling. Representation collapse as documented by Chi et al. [8] is an empirical pathology of trained MoE systems, and we have demonstrated only that one channel for it is closed by construction. Proposition 4 and S3 show that the other channel — data-side concentration — remains fully open, and is the one more likely to bite.

5.5 S5: at initialisation, the ablation arms are indistinguishable

Construction. For each ablation arm we measure the fraction of *dense* top-k attention mass that the sparse pattern retains ("recall"), against the fraction of causal cells the pattern admits ("density"). A random pattern of equal density has expected recall equal to its density, so the ratio is the pattern's selectivity. We run two input regimes: Gaussian noise, and sequences drawn from 32 recurring token embeddings — the latter because real text repeats token types across distant positions, which is the structure a content router is supposed to exploit.

input	variant	n	density	top-k recall	recall/density
gaussian	phasor	256	0.0909	0.2735	3.008
gaussian	phasor	1024	0.0464	0.2105	4.54
gaussian	a1_frozen_rotation	256	0.0907	0.2736	3.015
gaussian	a1_frozen_rotation	1024	0.0464	0.211	4.549
gaussian	a3_route_on_rope	256	0.0908	0.2738	3.015
gaussian	a3_route_on_rope	1024	0.0463	0.2107	4.547
gaussian	a4_pure_skip	256	0.0909	0.2735	3.008
gaussian	a4_pure_skip	1024	0.0464	0.2105	4.54
structured	phasor	256	0.0892	0.2691	3.017
structured	phasor	1024	0.0456	0.206	4.517
structured	a1_frozen_rotation	256	0.0911	0.2719	2.985
structured	a1_frozen_rotation	1024	0.0467	0.2086	4.469
structured	a3_route_on_rope	256	0.091	0.2743	3.013
structured	a3_route_on_rope	1024	0.0462	0.2158	4.672
structured	a4_pure_skip	256	0.0892	0.2691	3.017
structured	a4_pure_skip	1024	0.0456	0.206	4.517

Table 4: Pattern selectivity at initialisation ($d = 128$, 4 heads, window 8, $k = 32$). Angular bucketing beats a random pattern of equal density by a factor of three to five, but no arm is distinguishable from any other, in either input regime.

What the measurement shows. Two things, and the second is the useful one.

The cross-polytope partition is a genuinely good pattern prior: recall exceeds density by 2.5 to 4.7 times, and the margin *grows* with context (3.0× at $n = 256$, 4.5× at $n = 1024$ for the same window). Angular bucketing is doing real work relative to selecting the same number of keys at random.

But every arm does it equally well. Learned rotation, frozen random rotation, routing on content dimensions, routing on rotary dimensions, with and without residue summaries — all land within 0.011 of each other in recall at matched density, under both input regimes. Planting content structure in the input does not separate them either.

Constraint on how A1 and A3 can be tested

Pattern geometry at initialisation carries no information about which routing rule is better. Any difference between the ablation arms must be *created by training*, which means A1 (learned versus frozen rotation) and A3 (content versus rotary routing) cannot be settled by cheap pattern-quality measurement and require full training runs. It also means a reader should treat an initialisation-time pattern comparison — ours or anyone's — as uninformative about routing rules.

This is the clearest reason the fixed-hash null of Roller et al. [33] has to be taken seriously: before training, our learned router and Reformer's random one are the same pattern.

5.6 S6: the residue branch is a free hyperparameter the paper did not declare

bias at init	residue mass	inside P3's 2–25% band
0.0	0.6178	no
-2.0	0.1881	yes
-4.0	0.031	yes
-6.0	0.0043	no
-8.0	0.0006	no

Table 5: Fraction of attention mass taken by the residue branch at initialisation, against its logit bias ($d = 128$, 4 heads, $n = 512$, window 8). Prediction P3 assumes a 2–25% band; only two of the five settings reach it, and the default of zero is far outside.

With the residue logits entering the shared softmax at zero bias, the branch takes 62% of the attention mass before any training. Prediction P3 states that mass near one would indicate "routing has failed"; at initialisation it indicates nothing except that B well-conditioned logits compete against a sparse fine branch. The implementation now initialises the bias to -4, which starts the branch at 3%.

Two consequences. P3 must be a statement about the *converged* model and the paper must declare the initialisation, or the prediction is unfalsifiable because its outcome is settable. And the initialisation is a hyperparameter nobody has tuned, plausibly as consequential as anything else in ablation A4; it is now swept there.

5.7 S7: chunking is faithful only when the chunk matches the bucket

Construction. The $O(n^{1.5})$ complexity in Section 4.1 depends on sorting tokens by bucket and attending within fixed-size chunks, as Reformer does. That is not the same function as attending within buckets: a bucket larger than the chunk is split and loses intra-bucket pairs, and a smaller one is padded with a neighbouring bucket's tokens. We measure the gap on the *pattern*, which requires no fast kernel. We report

in-bucket recall (the fraction of true causal in-bucket pairs the chunked pattern keeps) and contamination (the fraction of the chunked pattern's pairs that are not in-bucket at all).

n	chunk c	$c/\bar{m}$	$H(\pi)/\log B$	in-bucket recall	contamination
256	32	4.00	0.978	0.8972	0.8811
256	64	8.00	0.978	0.948	0.9328
256	128	16.00	0.978	0.9777	0.9594
1024	32	1.00	0.992	0.8607	0.5634
1024	64	2.00	0.992	0.9588	0.7532
1024	128	4.00	0.992	0.9803	0.8699
4096	32	0.25	0.996	0.3779	0.2632
4096	64	0.50	0.996	0.6927	0.3074
4096	128	1.00	0.996	0.9265	0.5313

Table 6: Fidelity of bucket-sorted fixed-size chunking ($d = 128$, 4 heads, $B = 32$, window 1). $\bar{m} = n/B$ is the mean bucket size. Load entropy is near-perfect throughout, so the loss of fidelity is not an imbalance artefact.

What the measurement shows. At $n = 4096$ with chunk 32, the chunked pattern retains 38% of true in-bucket pairs. Load entropy in the same runs is 0.996, so this is not caused by unbalanced buckets — it is intrinsic. Figure 2(c) identifies the governing variable: plotted against $c/\bar{m}$, the three context lengths collapse onto a single curve. Fidelity above 90% requires $c \gtrsim \bar{m} = n/B$.

What this does to the complexity claim. Section 4.1 balanced the in-bucket term n^2/B against the residue term nB and obtained $B \asymp \sqrt{n}$, hence $O(n^{1.5})$. The chunk-fidelity requirement adds a constraint the analysis omitted: $c \gtrsim n/B$, so the in-bucket work per query is $\Omega(n/B)$ and the total is $\Omega(n^2/B)$ — which is $O(n^{1.5})$ only if B grows as $\sqrt{n}$. Under Definition 1, $B = 2d_c$ is the head's content width. So reaching the claimed operating point at context n requires

$$d_c \approx \tfrac{1}{2}\sqrt{n}, \tag{13}$$

that is, $d_c = 32$ at $n = 4096$ and $d_c = 128$ at $n = 65{,}536$. The first is ordinary; the second is a 128-dimensional content subspace per head on top of the positional dimensions, which is a real architectural cost that the complexity figure hides.

Qualification of the complexity claim

The $O(n^{1.5})$ figure is not wrong, but it is conditional on an operating point the paper did not state: the routing width must scale as $\sqrt{n}$, because $B = 2d_c$ and chunk fidelity requires $c \gtrsim n/B$. At fixed d_c the method is quadratic with constant $1/B$. Multi-round routing relaxes the constraint by unioning n_r independent partitions, at cost linear in n_r; whether that is a better trade than widening d_c is an open question we have not measured. Any wall-clock claim must report in-bucket recall alongside it, or it is reporting the speed of a different function from the one the method defines.

5.8 Toy-scale ablations, and a trap in how they are budgeted

The reference implementation applies routing as a mask over the full (n,n) score matrix, so it is $O(n^2)$ and says nothing about speed; what it can do is check that an arm trains. We run the arms on a synthetic copying

task — the second half of each sequence repeats the first half — which is sensitive to exactly the long-range dependency sparse attention can break, unlike random tokens, where every pattern scores the same. **The sweep is incomplete at the time of writing**; Table 7 lists the arms that have finished.

arm	params	best loss (seed 0 / 1)	density	$H(\pi)/\log B$
dense	393 856	3.870 / 3.434	1.000	–
phasor	394 384	4.160 / 4.158	0.086	0.738

Table 7: Toy scale, **sweep incomplete**: 2 layers, $d = 128$, 4 heads, context 256, vocabulary 64, 1000 steps, two seeds reported separately. These runs cannot support a claim about language modelling. The unigram floor is $\log 64 = 4.159$; PHASOR's density is 8.6% at window 8.

What the numbers say, read naively. Dense reaches 3.43 and 3.87; PHASOR sits at 4.160 and 4.158 — the unigram floor to three decimals, on both seeds. Read as a result, this says the sparse arm cannot do the task.

What the trajectories say. They do not support that reading. The loss curves are flat at the floor and then fall sharply: dense seed 1 leaves the floor at step 400 and reaches 3.43; dense seed 0 stays at the floor until step 900 and has only reached 3.87 when the budget ends. The task is learned by a phase transition whose *onset time* varies by more than a factor of two across seeds of an otherwise identical configuration. PHASOR's two seeds are both still flat at step 1000.

So the honest statement is narrow: within a budget in which the dense baseline barely completes its transition, PHASOR has not begun one. Whether it would at step 2000 or 5000 is unmeasured, and nothing here licenses the claim that it cannot.

A trap in budgeting sparse-attention comparisons

If the training budget is set so that the dense baseline is near its phase transition rather than well past it, every slower-onset arm reports at the floor and the comparison appears decisive when it is only truncated. Our own first reading of this table fell into it. Any budget for A1/A3/A4 must be long enough that *the dense arm has plateaued*, not merely long enough that it has started to learn — and the trajectories, not the final numbers, must be reported so a reader can tell the difference.

Taken with S5 — the arms are indistinguishable before training — the position is that A1 and A3 require training runs that are both long enough to pass the transition and replicated enough to resolve onset-time variance. Section 6 specifies that. Nothing cheaper will do, which is the single most expensive consequence of this report's findings.

6 Experimental Protocol, Revised

No language model has been trained. This section specifies the experiments that would settle what Section 5 could not, revised in four places where the studies showed the original design was wrong or uninformative. Revisions are marked. All result tables in Section 6.7 are placeholders and contain no data.

6.1 E0: the diagnostic that decides the RoPE contribution

Take a pretrained RoPE model. For each layer, extract key vectors both before and after the rotary embedding is applied, cluster each set on the sphere, and measure what the clusters encode.

Revised by S1, S2 and the synthetic run

The original specification led with $I(\text{cluster};\text{position})$. Two corrections. First, that quantity is *non-monotone* in context length — in a synthetic run with all confounds removed it rose to 0.38 at $n = 256$, fell to 0.13 at $n = 4096$, and was 0.16 at $n = 16{,}384$ — because at short context the phase correlates with position and at long context it scrambles, while the damage grows throughout. The quantity that falls monotonically is $I(\text{cluster};\text{token identity})$, from 1.00 without RoPE to 0.50 at $n = 16{,}384$, and it should be the headline. Second, E0 must report **effective bucket count**, because the failure S2 identifies is a capacity ceiling rather than uniform hashing, and mutual information does not see it.

A third point follows from S2 and is easy to get wrong: the ceiling is a property of *fixed-codebook argmax* routing. Spherical k-means fits its centroids to wherever the data went and does not exhibit it — in the synthetic run the effective cluster count stayed near 29 of 32 under RoPE while the argmax router collapsed to five. **E0 must therefore be run with both clustering rules**, or it will measure the wrong failure.

Specification. Models: Pythia-160m/410m or a comparable open RoPE model; the hook must capture keys on both sides of `apply_rotary_pos_emb`, and the two tensors must be asserted to differ before any number is trusted. Data: WikiText-103 or FineWeb-Edu. Clustering: spherical k-means with k-means++ seeding and three restarts, *and* fixed-codebook argmax, at $B \in \{16, 64, 256\}$. Contexts: 512 to the model's maximum. Report $I(\text{cluster};\text{token})$, $I(\text{cluster};\text{position})$ and effective bucket count, split by RoPE frequency tercile.

Falsification. If pre- and post-RoPE keys cluster comparably on real data, Proposition 2 has no practical force at the scales that matter and the RoPE-decoupling contribution should be withdrawn, not reformulated.

6.2 Models, data and matching

Two scales, trained from scratch: **S** (∼125M non-embedding parameters, 12 layers, $d = 768$, 12 heads) and **M** (∼350M, 24 layers, $d = 1024$, 16 heads). Data: FineWeb-Edu [30], with WikiText-103 [28] and PG-19 [31] for long-context evaluation. Token budgets follow compute-optimal ratios [18]. Contexts $\{1\text{k}, 4\text{k}, 16\text{k}, 32\text{k}\}$.

Added by S7

Remark 3 means the routing width cannot be held fixed across the context sweep without changing what is being measured: at $d_c = 32$ the method is at its claimed operating point at $n = 4096$ and is quadratic-with-small-constant at $n = 32{,}768$. Runs must therefore either scale d_c as $\sqrt{n}$, or hold d_c fixed and report that the complexity regime changes across the sweep. Reporting a single speed curve across contexts at fixed d_c would be misleading.

Every baseline is matched on total parameters, parameters active per token, training FLOPs and training tokens. Runs that cannot be matched on all four are reported separately and labelled. **Every table states what it is matched on**; the released harness records this in the run metadata so it cannot be left implicit.

6.3 Baselines

1. Dense transformer with FlashAttention [14] and full RoPE.
2. Dense transformer with partial RoPE at the same $\rho = d_{\mathrm{p}}/d_h$, which isolates the effect of the split from the effect of routing.
3. Reformer LSH attention [22] at matched bucket count and rounds.
4. Routing Transformer [34] at matched cluster count.
5. Sliding window with global tokens [4] at matched density.
6. Mixture-of-Depths [32] at matched capacity — the baseline for the fallback claim.
7. NSA [42] and/or MoBA [26] where a faithful reimplementation is feasible; if not, state so rather than omitting silently.

Baselines 1–6 are reachable in the released implementation by changing one flag, so A1, A3 and A4 are single-variable interventions rather than comparisons between separately written systems that differ in a dozen incidental ways.

6.4 Metrics

Validation perplexity; downstream zero-shot accuracy; wall-clock training step time and decode latency on a stated GPU; peak memory; KV read bandwidth per decode step; bucket-load entropy $H(\pi)/\log B$ over training; **effective bucket count** (added by S2); fraction of attention mass on residue summaries; recall of the sparse pattern against dense attention; and **in-bucket recall of the chunked kernel** (added by S7), without which a wall-clock number describes a different function from the one the method defines.

Added by the toy sweep

Report **loss trajectories, not only final numbers**, and run every arm until the dense baseline has *plateaued* rather than merely started to learn. On the toy task the onset of the learning transition varied by more than a factor of two across seeds of an identical configuration, and a budget truncated near that onset makes every slower arm report at the floor — which looks like a decisive negative result and is not one. Seeds must be sufficient to resolve onset-time variance, which is a different and larger requirement than resolving steady-state variance.

6.5 Ablations

6.6 Predictions

We commit in advance, so that what remains is falsifiable. P1 and P3 are restated from their original form in light of the studies.

P1′. A3 shows a perplexity gap favouring content-dimension routing that *widens* with context length, and the gap correlates with the effective-bucket-count gap rather than with mutual information against position. (Original P1 predicted a widening gap tracked by $I(\text{cluster};\text{position})$; S1 showed that metric is non-monotone.)

P2. A1 shows learned $\mathbf{R}$ beating random fixed $\mathbf{R}$ at equal B and n_r, with the gap largest at small B.

ID	Ablation	What it tests, and what we now know
A1	learned $\mathbf{R}$ vs. random fixed $\mathbf{R}$	Does learning the frame beat Reformer-style random hashing? S5 showed the two are indistinguishable at initialisation, so this requires training and cannot be shortcut. If there is no gap, the central claim fails [33].
A2	orthogonal $\mathbf{R}$ vs. unconstrained linear router	Is the constraint helping (stability, Prop. 1) or hurting (expressivity)?
A3	route on content dims vs. on RoPE'd dims	The decoupling claim. S1/S2 establish the mechanism; this establishes whether it costs perplexity. Also requires training, by S5.
A4	residue summaries vs. pure skip vs. fixed global tokens, $\times$ residue bias $\in \{0, -2, -4, -6\}$	The fallback claim. Bias sweep added by S6: the branch's share of mass is settable over two orders of magnitude, so the arm is not defined without it.
A5	B, n_r, w, ρ sweep, *with* chunk size c	Sensitivity. c added by S7: $c/\bar{m}$ is a correctness parameter, not a performance one.
A6′	*replaced:* encoder-side balancing and routing-input whitening, vs. router-side λ	**Respecified.** Proposition 4 shows router-side balancing cannot change the load. The question is whether whitening the routing input restores balance under anisotropic encoder output.
A7	bucket–POS/entity mutual information	Descriptive only: do buckets align with linguistic categories? A negative result does not falsify the method.
A8	Cayley vs. Householder(p)	Cost and expressivity of the orthogonal parametrisation.

Table 8: Ablations, revised. A1 and A3 remain the decisive ones; A6 has been replaced outright.

P3′. A4 shows residue summaries beating a pure skip at equal FLOPs, and in the *converged* model at residue bias -4 the attention mass on summaries lies between 2% and 25%. (Original P3 omitted both the convergence qualifier and the bias, which S6 showed makes the band settable and the prediction unfalsifiable.)

P4. A6′ shows whitening the routing input restoring load entropy under anisotropic encoder output, where router-side λ does not.

P5. Wall-clock gains appear only above a crossover length — we guess 8k–16k at scale S on current hardware — and PHASOR is *slower* than dense FlashAttention below it. Any such measurement is reported with in-bucket recall alongside.

P6. *New, from S7.* In-bucket recall above 90% requires $c \gtrsim n/B$, and holding d_c fixed while growing n produces a measurable perplexity degradation relative to scaling $d_c \propto \sqrt{n}$ at matched FLOPs.

6.7 Placeholder result tables

7 Positioning: what is new here

We summarise the claim structure explicitly, since the surrounding literature is dense and partial overlaps are easy to overstate.

Method	PPL (4k)	PPL (16k)	PPL (32k)	Step (ms)	Decode (ms/tok)
Dense + FlashAttention	[-]	[-]	[-]	[-]	[-]
Dense, partial RoPE	[-]	[-]	[-]	[-]	[-]
Reformer (LSH)	[-]	[-]	[-]	[-]	[-]
Routing Transformer	[-]	[-]	[-]	[-]	[-]
Longformer-style	[-]	[-]	[-]	[-]	[-]
Mixture-of-Depths	[-]	[-]	[-]	[-]	[-]
NSA / MoBA	[-]	[-]	[-]	[-]	[-]
PHASOR	[-]	[-]	[-]	[-]	[-]

Table 9: **Placeholder — contains no data.** Main comparison at scale S, FLOP- and token-matched, d_c scaled with context per Section 4.1.

ID	Variant	PPL	$H(\pi)/\log B$	eff. buckets	top-k recall
A1	random fixed **R**	[-]	[-]	[-]	[-]
A1	learned **R**	[-]	[-]	[-]	[-]
A2	unconstrained linear router	[-]	[-]	[-]	[-]
A3	routing on RoPE'd dims	[-]	[-]	[-]	[-]
A3	routing on content dims	[-]	[-]	[-]	[-]
A4	pure skip (MoD-style)	[-]	[-]	[-]	[-]
A4	residue summaries, bias -4	[-]	[-]	[-]	[-]
A6′	no whitening	[-]	[-]	[-]	[-]
A6′	routing-input whitening	[-]	[-]	[-]	[-]

Table 10: **Placeholder — contains no data.** Ablations at scale S, 16k context.

8 Limitations and Open Problems

The failure modes in Section 4.4 were written before the studies. This section states where the design stands now, in rough order of how likely each is to sink it.

1. The width–context coupling (Remark 3). The $O(n^{1.5})$ operating point requires $d_c \approx \sqrt{n}/2$, because $B = 2d_c$ and chunk fidelity requires $c \gtrsim n/B$. At fixed routing width the method is quadratic with a constant factor. This is the most consequential limitation and it is structural: it follows from the router's definition, not from an implementation choice. Multi-round routing is the obvious relaxation and is unmeasured. Decoupling B from d_c — for instance by composing several independent cross-polytope partitions of disjoint subspaces, which would multiply the effective codebook without widening any single rotation — is the design change we would try first, and it is not in this paper.

2. Load collapse, which is the encoder's to cause and to fix. Proposition 4 guarantees balance for isotropic routing inputs and guarantees nothing otherwise, and S3 shows the transition is sharp: by $\kappa = 4$ the load entropy is 0.10. Whether a trained encoder's key representations are close enough to isotropic in the routing subspace is an empirical question nobody has answered, and the answer decides whether the n^2/B accounting holds at all. If they are strongly anisotropic, whitening is the remedy and it has its own costs, which A6′ would measure.

3. Rigidity may cost accuracy. A fixed equiangular partition is less expressive than data-fit centroids. Routing Transformer's k-means allocates resolution where the data is dense; PHASOR cannot, except by

Component	Closest prior work	Status
Angular routing of tokens for sparse attention	Reformer [22]; Routing Transformer [34]	*Not new.* Established.
Cosine/normalised routing to fight collapse	X-MoE [8]	*Not new.*
Learned *orthogonal* frame with *fixed* orthonormal codebook, with the non-collapse guarantee of Prop. 1	Between Reformer (random frame) and Routing Transformer (free centroids); orthogonal parametrisations from [25, 29]	*New as a router design,* to our knowledge; the ingredients are standard.
Decoupling routing from RoPE, with Prop. 2	Partial RoPE [6]; decoupled RoPE in MLA [16] — both for other reasons	*The mechanism is standard; the routing-specific argument we have not seen stated.*
Residual carry of unrouted tokens	MoD [32]; MoE overflow [17]	*Not new.*
Making the router's residue *visible to the softmax* as per-query coarse summaries	NSA compression branch [42]; global tokens [4, 43]; sinks [41]; registers [15]	*Partly new:* prior coarse branches are routing-independent; ours is the router's complement, hence query-dependent.
Torus confinement as the explanation for the capacity ceiling (Prop. 3)	We are not aware of a prior statement; adjacent to Chiang and Yogatama [9] on rotary dimension inefficiency, which concerns the score rather than the partition	*New, to our knowledge.*
Load invariance under an orthogonal router with a fixed codebook (Prop. 4)	Balance-by-construction in BASE layers [24] achieves balance by a global solve, not by geometry	*New, to our knowledge,* and it cuts against our own ablation design.
Chunk-fidelity condition $c \gtrsim n/B$ for bucket-sorted sparse attention	Chunking is standard since Kitaev et al. [22]; we have not seen its fidelity measured or its effect on the complexity claim stated	*New as a measurement;* the mechanism is obvious once stated.

Table 11: Honest provenance of each component. The lower block is what the studies in Section 5 added.

moving the whole frame rigidly. If the data manifold is anisotropic this is a real disadvantage — and note it is the same condition that causes problem 2, so the two failure modes are correlated rather than independent. A2 measures the cost.

4. The fixed-hash null. Roller et al. [33] show random fixed hashing is a strong router, and S5 now shows that before training our learned router and a frozen random one produce patterns of identical quality. The entire value of the learned frame therefore rests on what training does to it, with no cheaper evidence available. If A1 shows no gap, the central claim of this paper is false and should be reported as such.

5. Semantic specialisation probably will not appear. The intuition motivating this line of work is that buckets will correspond to linguistically meaningful categories. Evidence from MoE models suggests expert specialisation tends to be syntactic and positional rather than semantic [20], and probing work shows linguistic structure is distributed across layers rather than concentrated [12, 39]. We rest no efficiency claim on interpretability of buckets; A7 measures it as a descriptive question and a negative result there does not falsify the method.

6. Hardware. Gather and scatter over sorted buckets have poor arithmetic intensity relative to dense FlashAttention [14]. Reformer and Routing Transformer both report that theoretical FLOP savings translate imperfectly into wall-clock, and NSA [42] exists largely because of this. Every table must report wall-clock, not only FLOPs — and now also in-bucket recall, since a fast kernel that computes a different function is not a speedup.

7. The residue branch is under-specified and over-free. D1 and D2 in Section 3.6 are choices we made without evidence, and S6 shows one of them spans two orders of magnitude of behaviour. The branch's contribution to the method is currently the least constrained part of the design.

8. What we have not measured at all. Decoding behaviour, including whether the running summaries drift relative to a full recomputation over long generations; interaction with grouped-query attention, where the routing decision would have to be shared across a key–value group; behaviour under a trained rather than randomly initialised encoder, which is the regime every claim in this paper actually concerns; and anything at all above toy scale.

9 Conclusion

We have described a sparse attention layer whose router is a learned orthogonal rotation acting on a fixed orthonormal codebook, which reads a RoPE-free content subspace, and whose excluded tokens remain visible to the softmax as coarse residue summaries. We gave a non-collapse property of the router's partition, an equidistribution argument for why angular routing should not read rotary dimensions, a torus-confinement result explaining what goes wrong when it does, an invariance result showing that the bucket load belongs to the data rather than to the router, and a complexity and cache analysis that avoids the two overclaims most common in this area — that expert routing of projections reduces the quadratic term, and that sparse attention reduces KV cache capacity. Neither is true.

The more useful output of this work is probably the four corrections in Section 5, three of which were corrections to our own claims. Routing on rotary dimensions does not degrade toward uniform hashing but toward a capacity ceiling, which is worse and gets relatively worse as the codebook grows. A balancing loss on the router cannot work, which killed an ablation we had already designed. The chunking that the complexity figure depends on is faithful only under a condition the figure omits, which couples routing width to context length. And at initialisation every ablation arm produces an identical pattern, which means the remaining claims cannot be tested cheaply.

We have trained no language model, and we think a proposal is more useful stated that way than dressed in results it does not have. What the released code does provide is the reference implementation, a test suite that checks causality by autograd rather than by inspection, and the studies above. The cheapest decisive experiment is still E0, which requires no training; the revised version in Section 6.1 measures two quantities the original would have missed. We would run it before anything else.

Reproducibility. The implementation, tests and studies are released at `https://github.com/Alessio2405/phasor`. Running `pytest tests/` executes 128 tests covering orthogonality under optimisation, causality of every ablation arm by autograd, agreement between the training-time and decoding-time summary paths, and the numerical counterparts of each proposition. The following regenerates every table and figure in Section 5 from scratch:

```
python experiments/numerical_studies.py && python experiments/make_tables.py
```

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A Orthogonal parametrisations

Cayley. For skew-symmetric $A \in \mathbb{R}^{d_c \times d_c}$ ($A = -A^\top$, $d_c(d_c-1)/2$ free parameters), $\mathbf{R} = (\mathbf{I}-A)(\mathbf{I}+A)^{-1}$ is orthogonal with $\det \mathbf{R} = 1$, and the map is a diffeomorphism onto $SO(d_c)$ minus the eigenvalue -1 locus. Cost $O(d_c^3)$ per update, amortised over the batch.

Householder products. $\mathbf{R} = \prod_{t=1}^{p}(\mathbf{I} - 2\mathbf{h}_t\mathbf{h}_t^\top/\|\mathbf{h}_t\|^2)$ is orthogonal for any $\{\mathbf{h}_t\}$, with $\det \mathbf{R} = (-1)^p$. Taking $p = d_c$ spans $O(d_c)$; taking $p < d_c$ gives a controllable low-cost subfamily at $O(pd_c)$ per application [29]. This is our default for $d_c \geq 64$.

Exponential map. $\mathbf{R} = \exp(A)$ with trivialisation [3, 25]; equivalent guarantees, higher constant.

In all three cases, orthogonality holds *exactly* at every point of training, not approximately via a penalty — which is what Proposition 1 requires.

B Causal running summaries

For decoding we maintain per-bucket counts c_b and sums $\mathbf{S}_b^k, \mathbf{S}_b^v$. On emitting token j with bucket b: $c_b \mathrel{+}= 1$, $\mathbf{S}_b^k \mathrel{+}= \mathbf{k}_j$, $\mathbf{S}_b^v \mathrel{+}= \mathbf{v}_j$, and $\bar{\mathbf{k}}_b = \mathbf{S}_b^k / c_b$. State is $O(B(d_h + d_v))$, independent of n. During training the same quantities are computed with a causal segmented cumulative sum over the bucket-sorted order, so the summaries are exact and require no extra pass.

Note that $\bar{\mathbf{k}}_b$ is a mean of keys, so $\mathbf{q}^\top \bar{\mathbf{k}}_b$ equals the *mean score* of bucket b, not the max. This makes the residue branch a conservative (low-variance, potentially over-smoothed) estimate of what the bucket contains; a max-pooled or learned-pooled variant is an obvious alternative and is included in A4's extended form.

C Reproducibility checklist

- Reference implementation: https://github.com/Alessio2405/phasor; built on a nanoGPT-scale training harness.
- All runs report seed, exact token count, hardware, and wall-clock alongside FLOPs.
- Every table states whether entries are FLOP-matched, parameter-matched, or neither.
- Negative results (P1–P5 failures) are reported in the same table as positive ones.